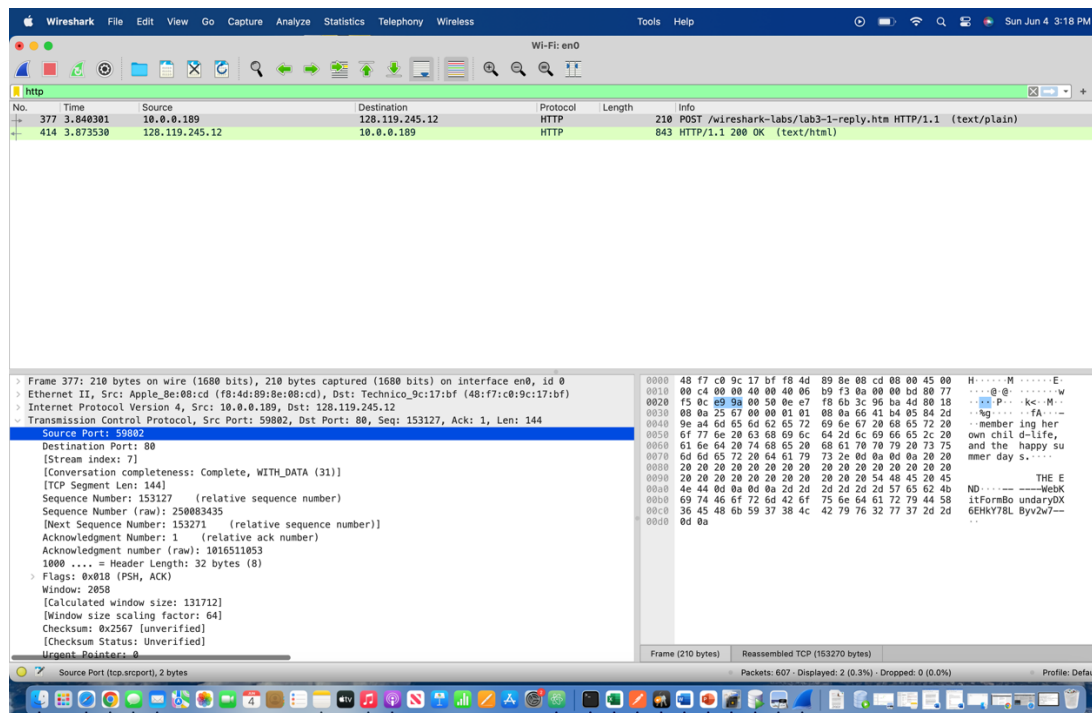


- 1) What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window" (refer to Figure 2 in the "Getting Started with Wireshark" Lab if you're uncertain about the Wireshark windows).

Source IP: 10.0.0.189

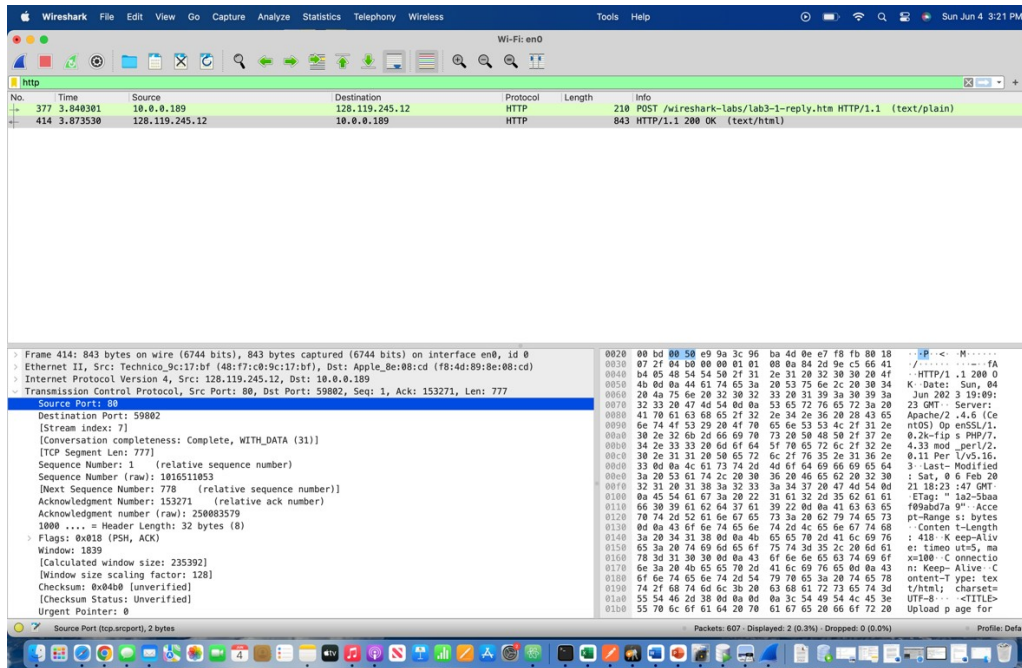
Source Port: 59802



- 2) What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

IP Address: 128.119.245.12

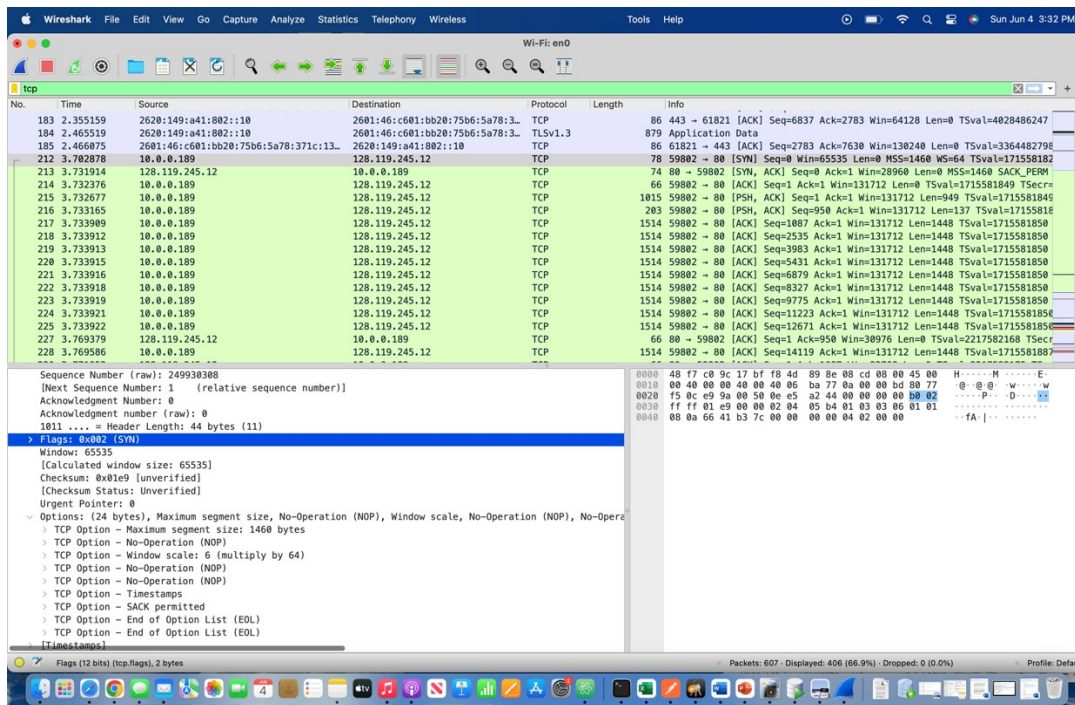
Port: 80



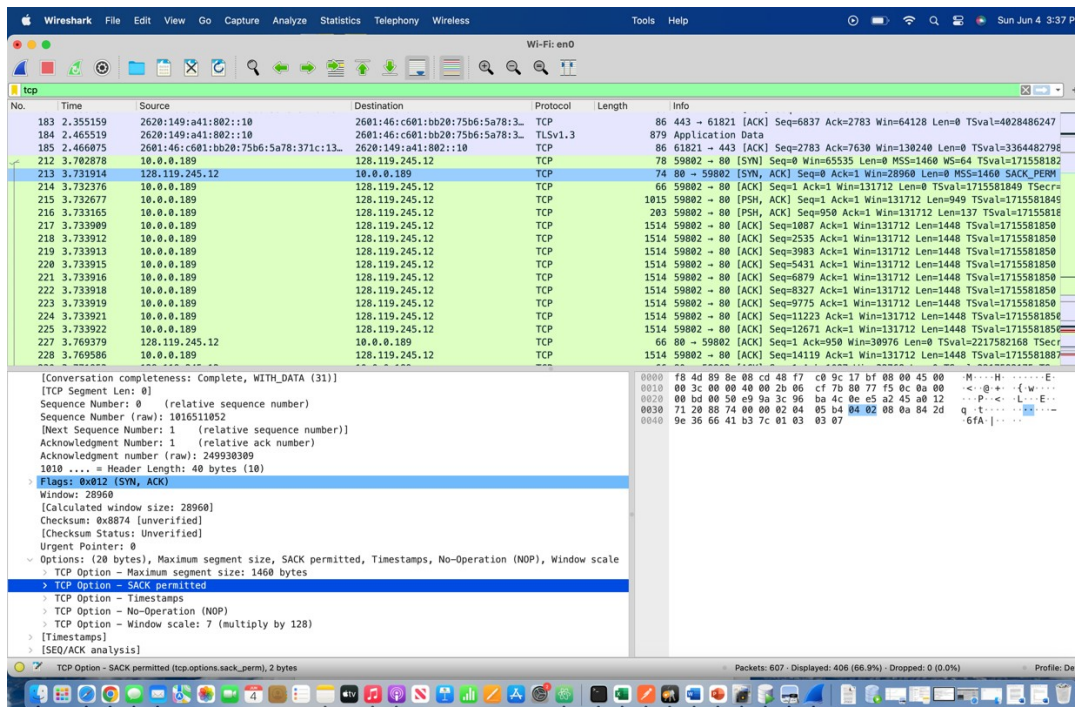
3) What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? (Note: this is the “raw” sequence number carried in the TCP segment itself; it is NOT the packet # in the “No.” column in the Wireshark window. Remember there is no such thing as a “packet number” in TCP or UDP; as you know, there are sequence numbers in TCP and that’s what we’re after here. Also note that this is not the relative sequence number with respect to the starting sequence number of this TCP session.). What is it in this TCP segment that identifies the segment as a SYN segment? Will the TCP receiver in this session be able to use Selective Acknowledgments (allowing TCP to function a bit more like a “selective repeat “receiver, see section 3.4.5 in the text)?

Sequence Number (raw): 249930308

In a TCP segment SYN(synchronize) flag is set to identify the segment as SYN segment.

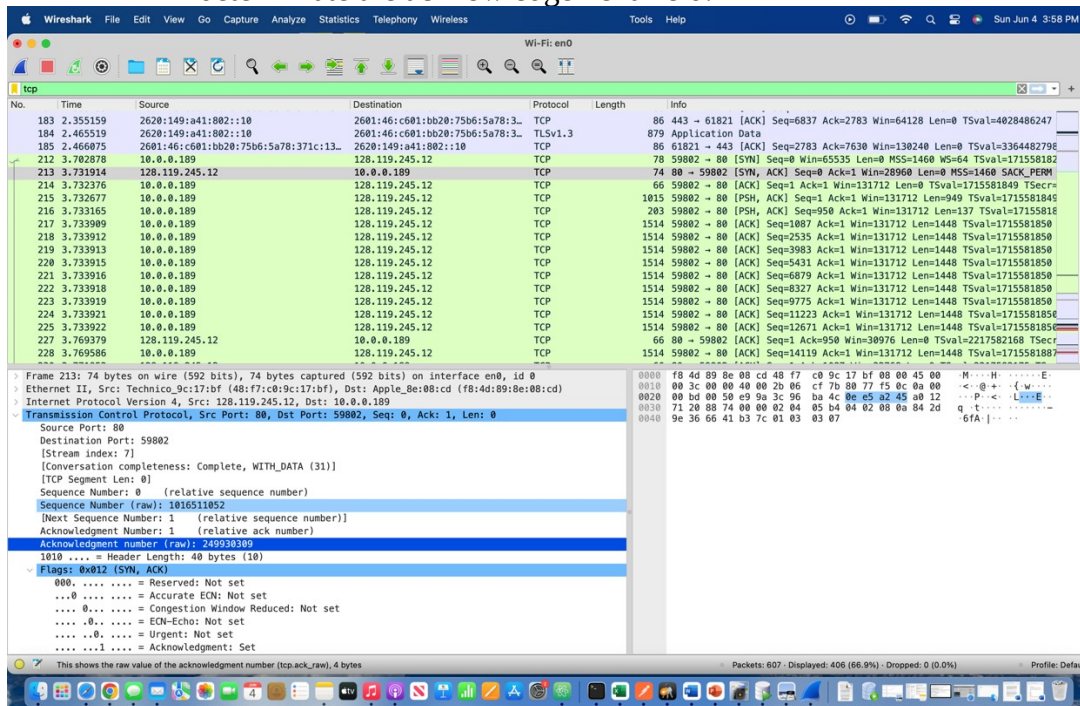


TCP receiver in this session be able to use Selective Acknowledgements.



- 4) What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is it in the segment that identifies the segment as a SYNACK segment? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?

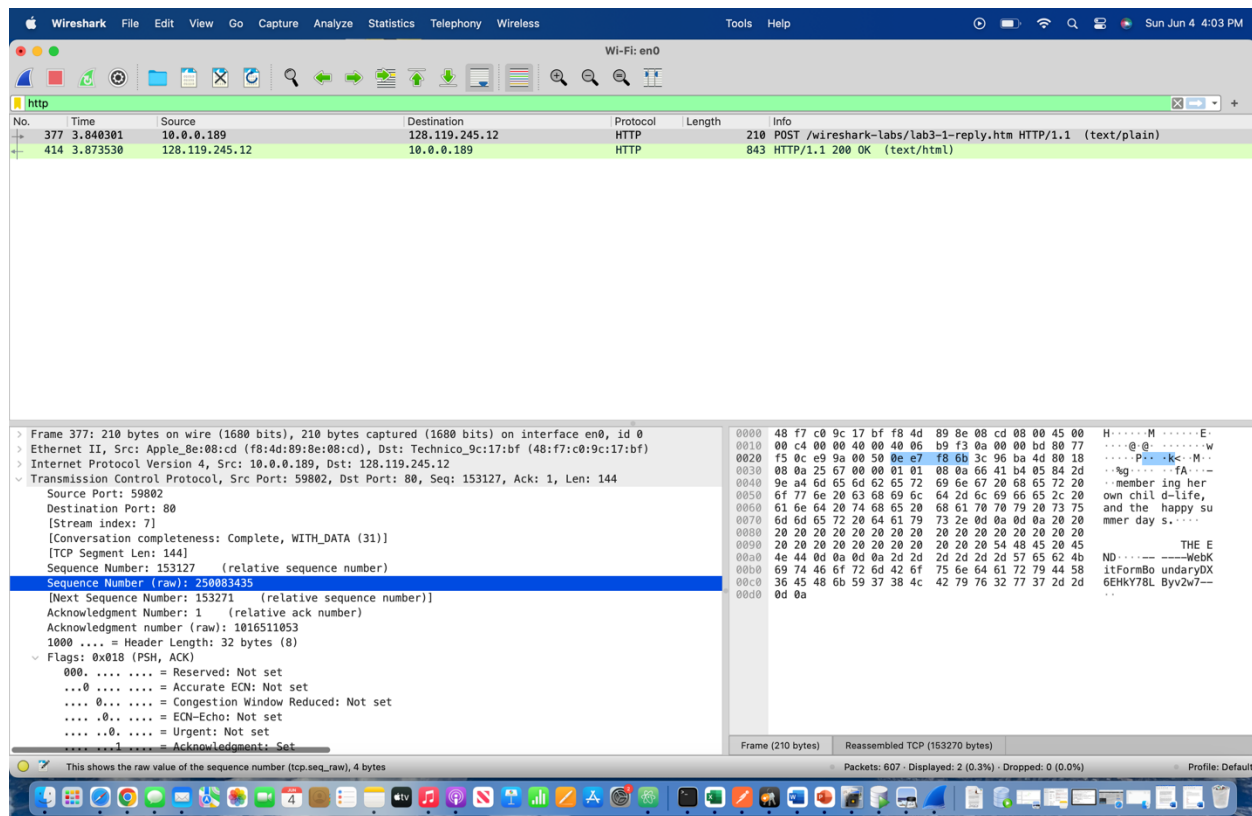
- Sequence Number (raw): 1016511052
- SACK permitted – identifies the segment as SYNACK
- Acknowledgment number: 249930309
- Gaia.cs.umass.edu determines by incrementing the sequence number by 1 to determine the acknowledgement field.



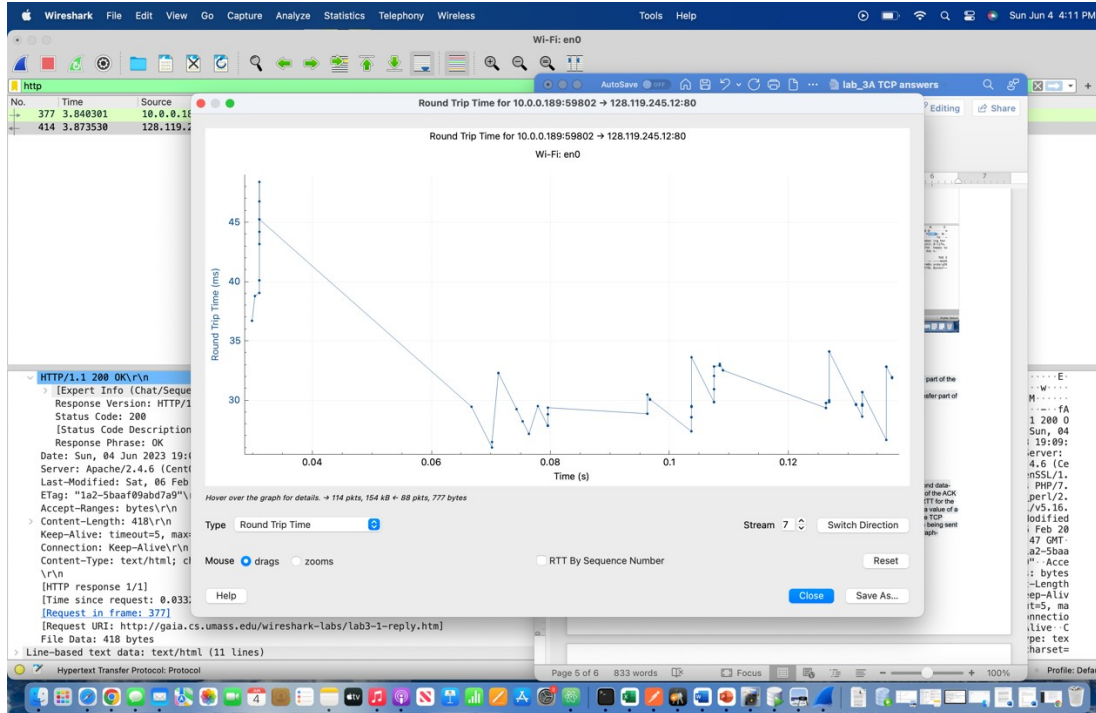
- 5) What is the sequence number of the TCP segment containing the header of the HTTP POST command? Note that in order to find the POST message header, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with the ASCII text "POST" within its DATA field. How many bytes of data are contained in the payload (data) field of this TCP segment? Did all of the data in the transferred file alice.txt fit into this single segment?

Sequence number: 250083435

949 bytes of data are contained in the payload field of this TCP segment. No not all the data in the transferred file alice.txt fit into this single segment.



- 6) Consider the TCP segment containing the HTTP “POST” as the first segment in the data transfer part of the TCP connection.
 - a. At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?
 - o Time: 3.840301
 - b. At what time was the ACK for this first data-containing segment received?
 - o Time: 3.873530
 - c. What is the RTT for this first data-containing segment?
 - o 0.033229
 - d. What is the RTT value the second data-carrying TCP segment and its ACK?
 - o 0.0254
 - e. What is the Estimated value (see Section 3.5.3, in the text) after the ACK for the second data-carrying segment is received? Assume that in making this calculation after the received of the ACK for the second segment, that the initial value of Estimated is equal to the measured RTT for the first segment, and then is computed using the Estimated equation on page 242, and a value of $\alpha = 0.125$. Note: Wireshark has a nice feature that allows you to plot the RTT for each of the TCP segments sent. Select a TCP segment in the “listing of captured packets” window that is being sent from the client to the gaia.cs.umass.edu server. Then select: Statistics->TCP Stream Graph->Round Trip Time Graph.

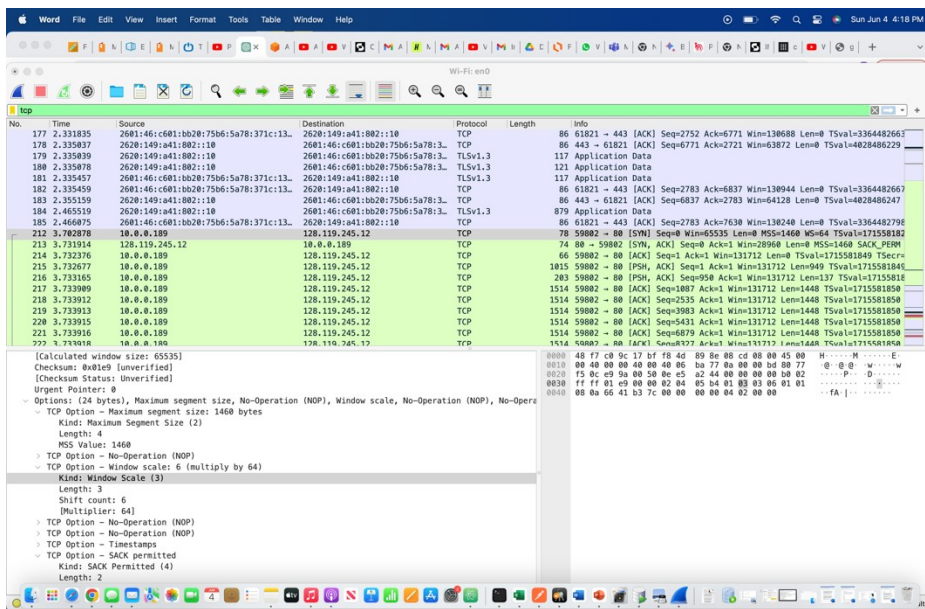


- 7) What is the length (header plus payload) of each of the first four data-carrying TCP segments?

Length: 144

- 8) What is the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments? Does the lack of receiver buffer space ever throttle the sender for these first four data-carrying segments?

Windows scale 6 (multiply by 64). Kind: Window scale (3), length: 3, shift count 6



- 9) Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?

There are no retransmitted segments all the data is in encoded format in the trace file. I have checked the t FRAM payload for this.

- 10) How much data does the receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu? Can you identify cases where the receiver is ACKing every other received segment (see Table 3.2 in the text) among these first ten data-carrying segments?

32 Bytes of data. I don't see any other cases where the receiver is ACKing every other received segment.

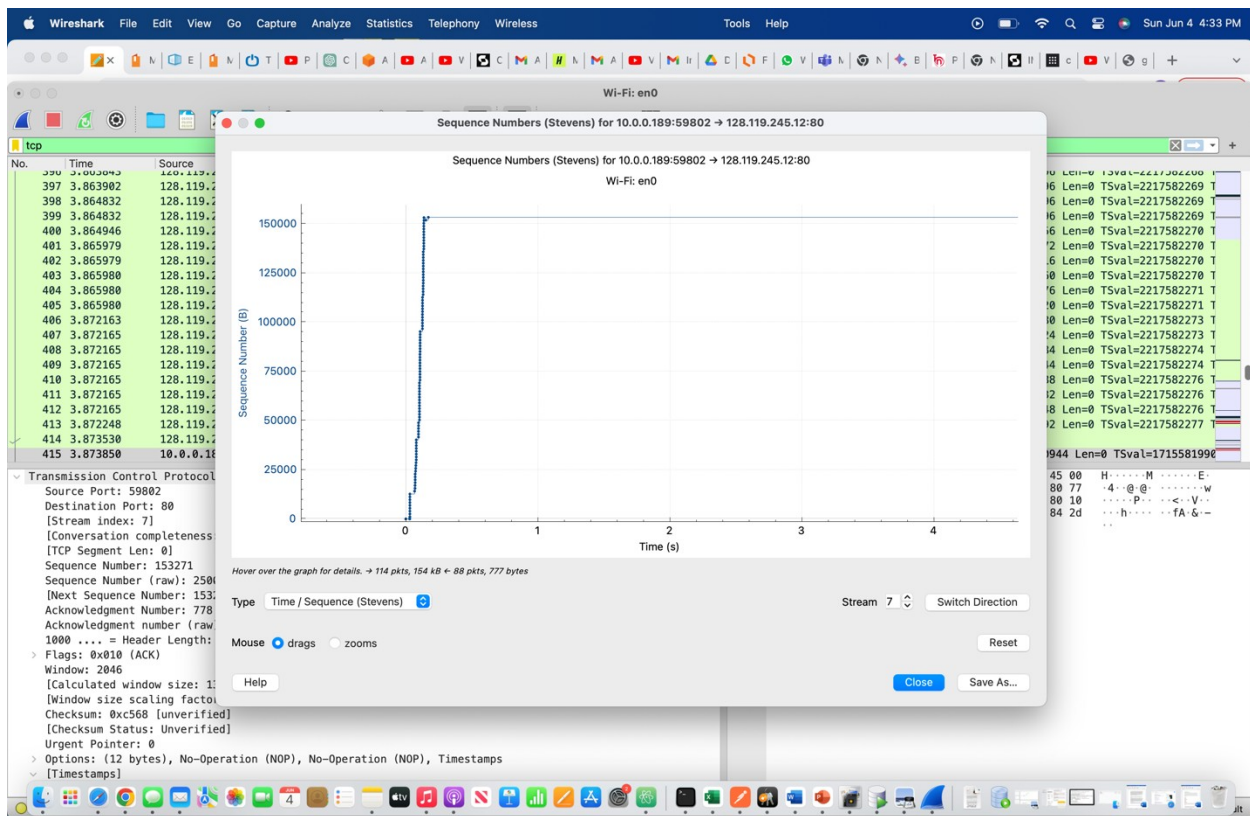
- 11) What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.

32 bytes. I have filtered on all the communication that needs to be considered and calculated the start and end time to find out the total amount of time it took to transfer the data and divided the total amount data transferred with the amount of time it took.

Throughput = Total bytes transferred / Time duration

- 12) Use the Time-Sequence-Graph(Stevens) plotting tool to view the sequence number versus time plot of segments being sent from the client to thegaia.cs.umass.edu server. Consider the “fleets” of packets sent around $t = 0.025$, $t = 0.053$, $t = 0.082$ and $t = 0.1$. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase. Figure 6 shows a slightly different view of this data.

Congestion avoidance phase.



13) These “fleets” of segments appear to have some periodicity. What can you say about the period?

For every milli second (~) there are 25000+ sequence numbers being transferred.

14. Answer each of two questions above for the trace that you have gathered when you transferred a file from your computer to gaia.cs.umass.edu

